From observations to freshness

Research definitions for one chosen identity and history

For a supplied finite event set E, let s_e be an event's sequence, p_e its publication time, and a_e its evidence-availability time. Only events with both times known and already reached contribute:

$$E_t = \{e \in E \mid p_e \text{ and } a_e \text{ known},\ p_e \le t,\ a_e \le t\}, \qquad F_E(t) = \max_{e \in E_t} s_e \quad (E_t \neq \varnothing).$$

The empty set yields *unknown*, not sequence zero. If the chain projects sequence c, the supported observed lag is

$$L(c, F) = \begin{cases} F - c & c \le F, \\ \text{ahead of observation } (c, F) & c > F. \end{cases}$$

Missing projection, unknown observations and unsupported observations have separate results. Example: $F = 3$, $c = 1$ gives $L = 2$ events. Zero observed lag does not rule out hidden events.

For one event with publication time p_e and inclusion time t_e,

$$D(e, t_e) = t_e - p_e \qquad \text{when } p_e \text{ is known and } p_e \le t_e.$$

Otherwise delay is undefined. Event counts and time are different units. Consumer maturity is another property:

$$\text{eligible}(s, t) \quad \not\Rightarrow \quad L(s, F_E(t)) = 0.$$

The draft exhibits an eligible checkpoint three events behind its observation.

Mirror obligation. For registry r and revocation identifier x, coverage means that r exists and its stored revoked set contains x. Completeness quantifies over valid anchored revocation seal walks:

$$\text{valid anchored revocation walk for } x \text{ in } r \implies \text{Covered}(r, x).$$

This is a proposed coverage requirement, not a completeness result established by the checkpoint metric. The draft proves no composition or independence of the two models.

Lean map: `frontierFrom`, `sequenceLag`, `projectionDelay`, `eligible`, `RevocationCovered`, `MirrorComplete`. Observations and their authenticity/completeness remain supplied research inputs.

Three conditional research claims

All three Lean statements are intentionally unproved

Accounting definition. For one agent and a fixed horizon,

$$U = R + P_{\text{returned}} + P_{\text{terminal}} - P_{\text{committed}} - F_{\text{sponsor}} - C.$$

C sums monitoring, success, failure, racing, locked-capital and manufacturing costs. Principal returned against equal principal committed is not profit. Units, costs, control of addresses and terminal valuation are research choices.

C1: submit versus stop. Restrict to a lawful keep rotation without refund change, with both choices feasible. Its account has no capital input:

$$U_{\text{submit}} = r - C, \qquad U_{\text{stop}} = 0, \qquad r > C \implies U_{\text{submit}} > U_{\text{stop}}.$$

r is the actual addressed rotation premium. Example: $10 - 3 = 7$; with costs 12, utility is -2. This comparison proves neither equilibrium nor timely submission. Lean: `C1_submit_over_stop`.

C2: compose separate delay bounds. For a lawful included event with ordered publication, observation, submission and inclusion times,

$$t_o \le t_p + \delta_o, \quad t_s \le t_o + \delta_s, \quad t_i \le t_s + \delta_i \quad \implies \quad t_i - t_p \le \delta_o + \delta_s + \delta_i.$$

Each bound needs its own justification. Inclusion is already a premise; this is not a liveness guarantee. Lean: `C2_delay_composition`.

C3: finite funded service. For an actual trace containing only rotations and pool top-ups, fixed premium $P > 0$, initial pool Q_0, total added funding T, and budget B,

$$Q_0 + T \le B \quad \implies \quad \sum_j r_j \le B \quad \wedge \quad N_{\text{paid}} P \le B.$$

This bounds premium spending and paid rotations, not all costs or unpaid progress. Example: $Q_0 = 20$, $P = 10$, $T = 0$ supports two paid updates. Lean: `C3_funded_work_bound`.

Discussion target. Choose the service metric, justify observations and costs, model competition and funding renewal, then strengthen or replace these statements. The Lean types are authoritative for their exact premises; these formulas are their reading guide.